

Exploratory Data Analysis (EDA) — Project Report

CodeAlpha Internship | Data Analytics Track | Task 2

About This Task

This task is about studying a dataset in detail before doing any deep analysis. This process is called **Exploratory Data Analysis (EDA)**. The goal is to understand what the data contains, find patterns, and present insights visually through charts.

What I Did

I analyzed a **Fashion Products dataset** to understand its structure, find interesting patterns, and create visual charts that explain the data clearly.

How I Did It

- I downloaded a fashion products dataset from **Kaggle**, which contains information like product category, gender, color, season, and year.
- I checked the basic details of the dataset — how many rows and columns it has, and what type of information each column holds.
- I checked for any **missing or incomplete data** in the dataset.
- I explored the data by asking simple, meaningful questions, such as: Which gender has the most products? What are the most common product categories? Which colors are most popular? How are products distributed across different seasons? How many products were added each year?
- I used **Pandas** to perform the analysis and **Matplotlib** and **Seaborn** to create visual charts (bar charts, pie charts, and a heatmap) that represent these patterns clearly.

What I Learned

- How to explore a brand-new dataset step by step before jumping into deeper analysis.
- How to detect missing data and understand how it can affect results.
- How to turn raw numbers into clear, easy-to-understand visual insights.
- How different chart types (pie, bar, heatmap, line) are useful for different kinds of data.
- How visualizations make it much easier to spot trends and patterns compared to just looking at numbers in a table.

Tools & Libraries Used

Tool	Purpose
Python	Programming language used
Pandas	To load, clean, and analyze the dataset

Matplotlib	To create basic charts and graphs
Seaborn	To create more advanced and styled visualizations

Files in This Repository

- **eda_analysis.py** — Python code used for the analysis
- **styles.csv** — Dataset used for analysis
- **Graph images (.png files)** — Visual charts generated from the analysis (gender distribution, top categories, popular colors, season trends, and more)

Submitted as part of the CodeAlpha Data Analytics Internship.